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Department of Electrical and Electronics Engineering

AGROVOLTAIC CANOPY FOR PROTECTION AND SUSTAINABLE FARMING

Review 0: prototype demonstration, six-sensor smart tilt controller (computer simulation)

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Problem and Need

Land, water and income are in conflict for small and marginal farmers

Water scarcity and cost

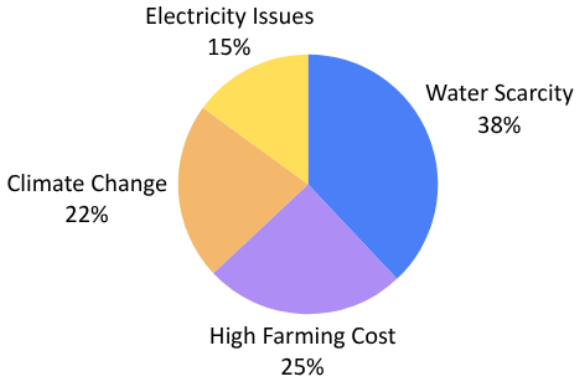
38 % of surveyed farmers named water scarcity and 25 % high farming cost as the top problem.

Solar takes farmland

Ground-mounted solar removes land from cultivation.

Fixed agrivoltaics cannot adapt

They cannot react to sun, cloud, heat, wind or rain, so they waste energy or shade the crop.





Field survey, Viluppuram district: 38 % water, 25 % cost, 22 % climate, 15 % electricity

The need

A low-cost canopy that chooses the panel angle from sensors, so that the crop gets the light it needs and the panels still make electricity, harvests rain for irrigation and stays safe in storms.

Review 0 question: can six sensors choose the angle accurately, and can we show it before building hardware?

Objectives and What Review 0 Delivers

Everything runs on the computer: no board, no wiring

Objectives of Review 0

1 Simulate six sensors from real, named datasheets, each adjustable by hand

2 Decide the panel angle from the sensors: safety, then water, then energy and crop light

3 Show the result live in a MATLAB dashboard with plain-word explanations

4 Measure the accuracy of the co-optimisation against the best possible angle

5 Prove every number: datasheets, NASA POWER and PVGIS data, tests, checksums

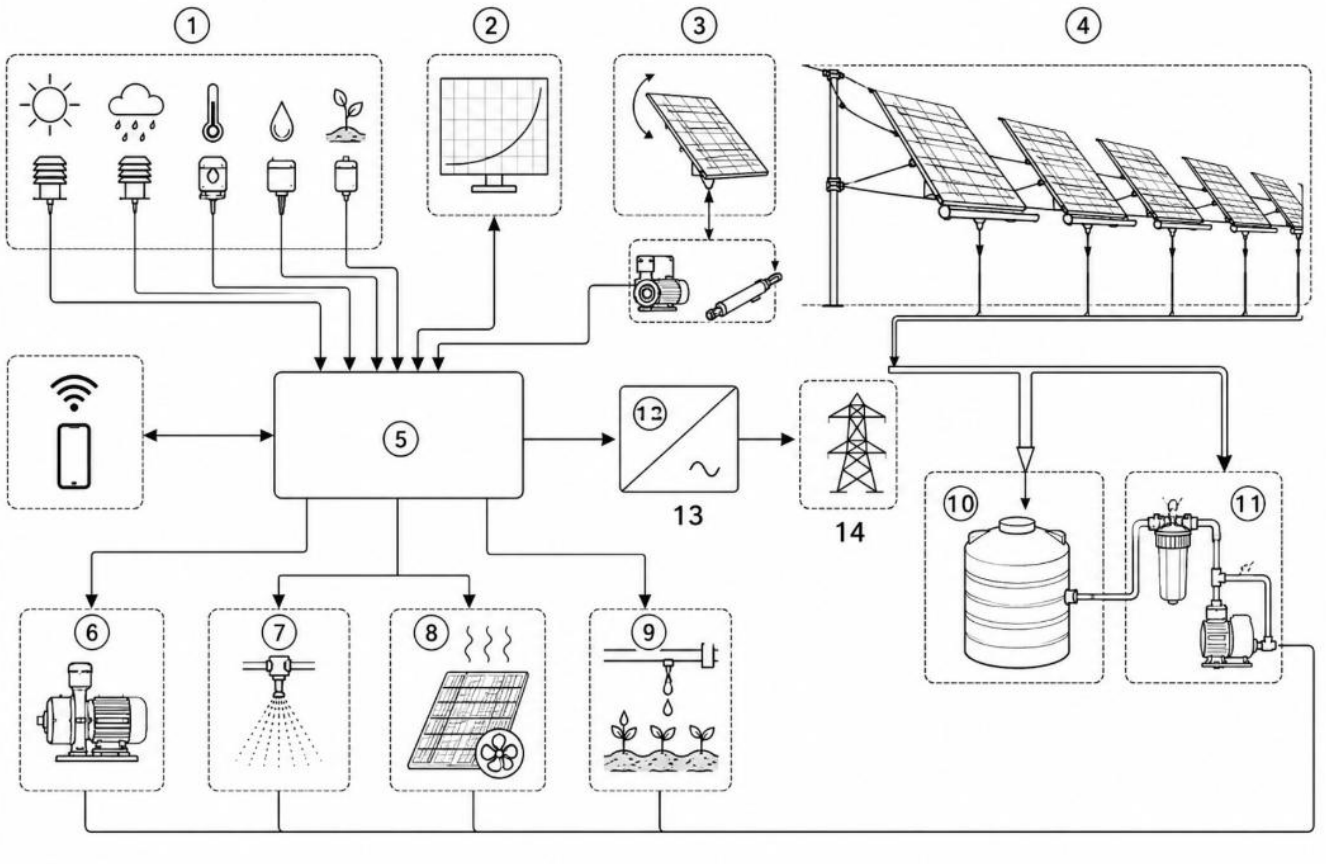
6 Estimate the cost if the sensors were real

Review rubric	Marks	Where in this deck
Prototype demonstration	20	slides 8, 9, 10
Testing and validation evidence	10	slides 11, 12, 13
Feasibility	5	slides 14, 15
SDG and cost	5	slides 14, 15
Time management	5	slide 16
Questions	5	slide 17

Honest status: a validated computer simulation. The sensor readings are generated or typed, not measured. No field trial yet.

The Idea and the System

Sensors in, best angle out: electricity for sale, light for the crop, rain for the tank



Site: TCE Madurai (9.88 N, 78.08 E). Panels: 8 x 550 Wp. Tilt range 15-60 deg; rain 3 deg; storm 90 deg.

How it works

1 Six sensors read sun, plant light, air, soil, rain, wind

2 The controller picks the tilt every step

3 A motor and wire tilt all panels together

Why it is new

The angle balances electricity and crop light (half and half, adjustable).

Plant stress (hot, dry air or soil) is used as an input.

Every decision is explained in plain words.

The Six Sensors: Simulated After Real Devices

Each reading follows the device datasheet; the datasheet files are downloaded as proof

#	Reading	Real device simulated	Range	Accuracy	Output signal
1	Sunlight	Apogee SP-110 pyranometer	0-2000 W/m2	+-5 %	0.20 mV per W/m2
2	Plant light (PAR)	Apogee SQ-110 quantum sensor	0-4000 umol/m2/s	+-5 %	0.20 mV per umol
3	Air temperature	DHT11	0-50 deg C	+-2 deg C	digital
4	Air humidity	DHT11 (same chip)	20-90 % RH	+-5 % RH	digital
5	Soil moisture	DFRobot SEN0193 v1.2	2.5 V dry to 1.2 V wet	needs dry/wet calibration	analog 1.2-2.5 V
6	Rain	YL-83 / FC-37 + LM393	wet / dry	trimmer threshold	digital, LOW = wet
7	Wind speed	Adafruit 1733 anemometer	0.5-32.4 m/s	+-1 m/s worst case	analog 0.4-2.0 V

How a simulated reading works: the slider is the physical quantity and the dashboard also shows the signal the real device would output (837 W/m2 on the SP-110 is 167 mV; 20 m/s on the Adafruit 1733 is 1.39 V). Every sensor can be set by hand.

The Solar Panels: 8 x JinkoSolar Tiger Pro 550 Wp

Real module, values read from the official datasheet we downloaded

Item	Value
Model	JKM550M-72HL4-V (mono)
Maximum power (STC)	550 Wp
Vmp / Imp	40.90 V / 13.45 A
Voc / Isc	49.62 V / 14.03 A
Efficiency	21.33 %
Size / weight	2274 x 1134 x 35 mm / 28.9 kg
Our array	8 panels = 4.4 kWp

The tilt controller moves all panels together by one wire. 40.90 V x 13.45 A = 550 W.

The array: 2 rows x 4 modules (landscape)

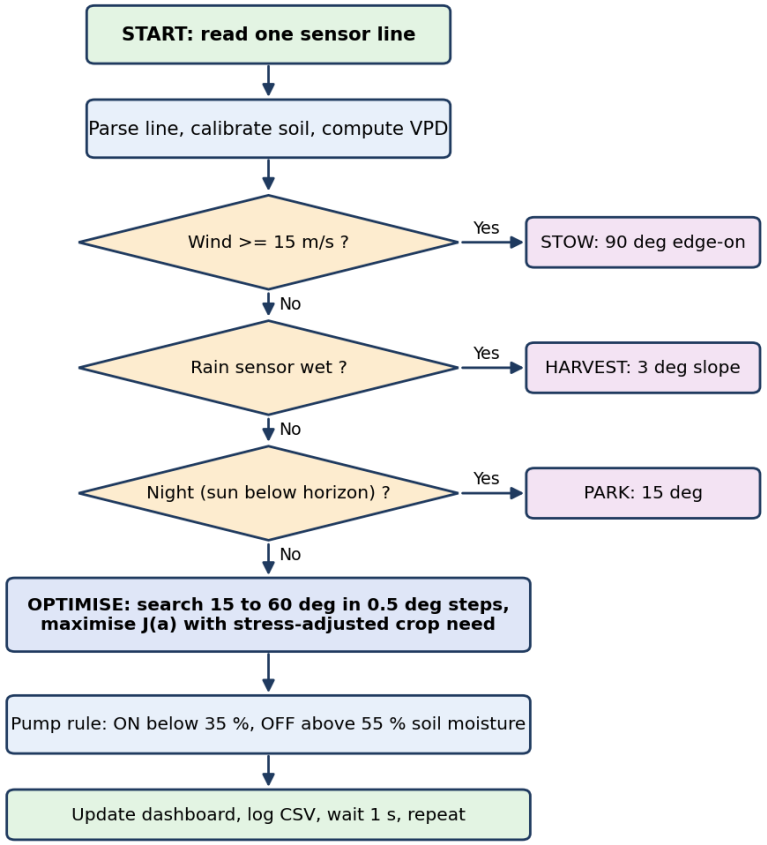


Series string voltage at maximum power: 4 x 40.9 = 163.6 V per row

Total 8 x 550 = 4,400 Wp

How the Controller Decides

Safety first, then water, then energy and crop light



Priority rules

1. **Wind 15 m/s or more:** stow edge-on at 90 deg
2. **Rain sensor wet:** 3 deg slope, rain runs to the tank
3. **Night:** park at 15 deg

Otherwise: co-optimise

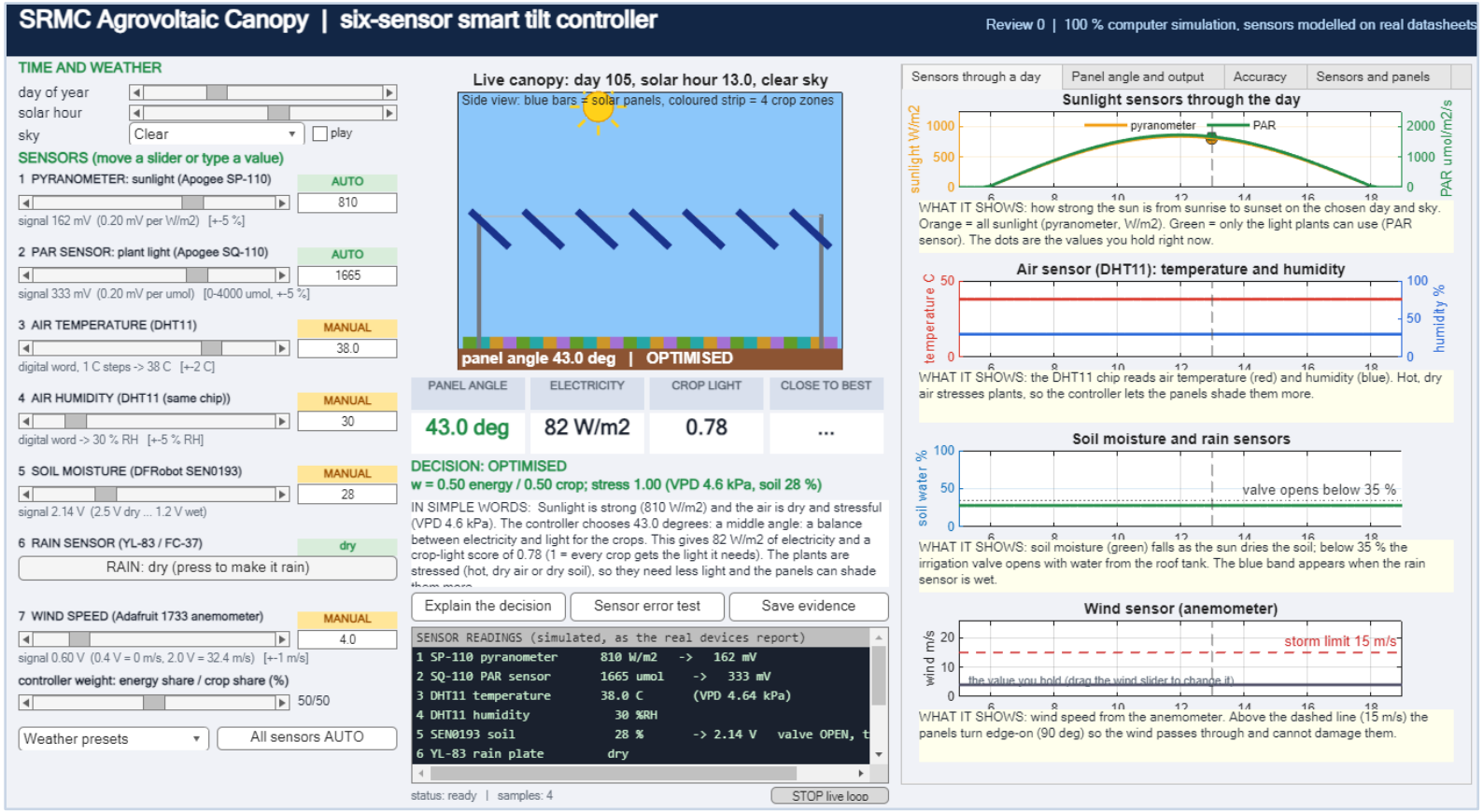
Try every angle from 15 to 60 deg (0.5 deg steps) and pick the best of

J(a) = w x Electricity(a) + (1 - w) x CropLight(a)

w = 0.5 by default (slider). Hot, dry air or dry soil lowers the light the crop needs, so the panels give more shade. Pump on below 35 % soil water, off above 55 %.

Prototype: the MATLAB Dashboard

Seven readings from six devices, all adjustable by hand, with the decision explained in words



What the judges can do

Move a slider or type a value for any sensor (MANUAL tag)

Press a weather preset: clear noon, cloud, hot dry, rain, storm, night

Press RAIN: panels go flat

Set wind to 20 m/s: panels stow at 90 deg

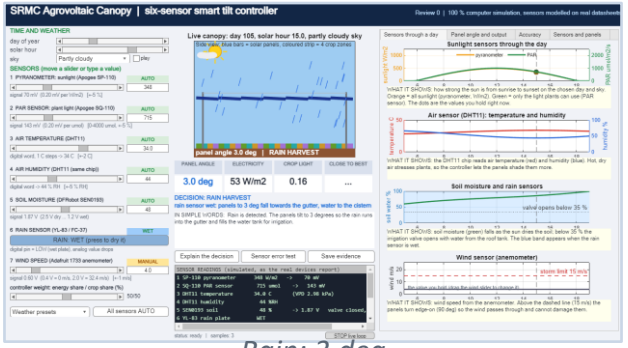
Read the decision in plain words

Open the four graph tabs

Live Demo: Change a Sensor, the Panels React

Five steps, about two minutes

1	**Clear noon** preset: panels settle at 43 deg for the best mix
2	**Drag sunlight or PAR** the angle changes with the light
3	**Hot, dry preset** crop is stressed: more shade
4	**Press RAIN** panels flat at 3 deg, tank fills
5	**Wind to 20 m/s** stow edge-on at 90 deg



Rain: 3 deg



Wind 20 m/s: 90 deg

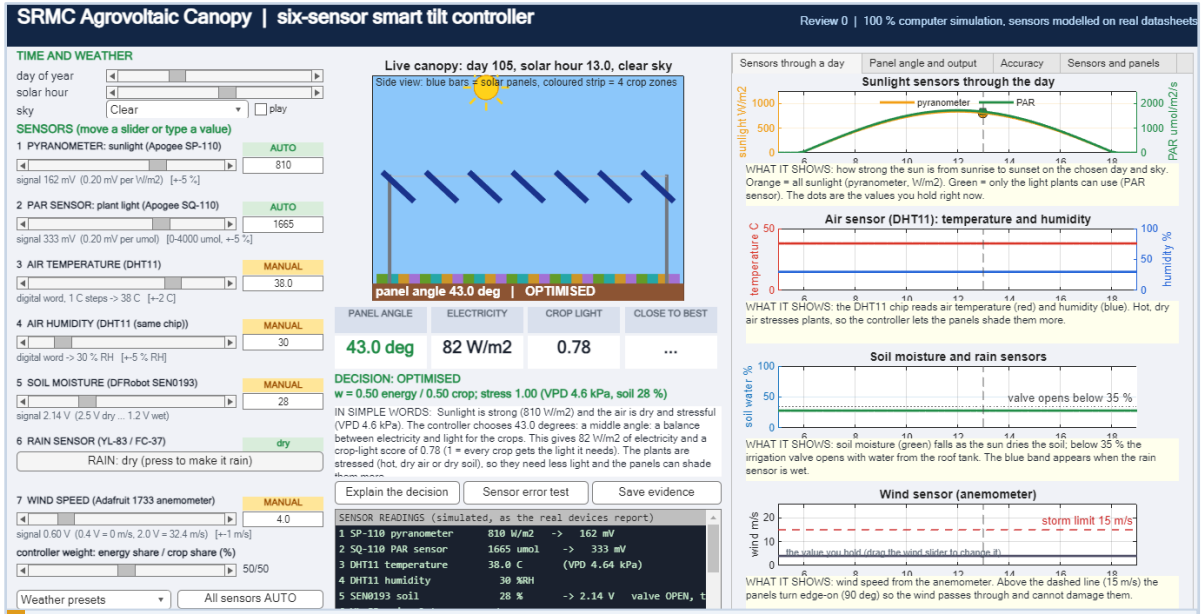
Each graph says what it shows

Under every graph there is a one-line explanation in plain words, so a judge can read the dashboard without help.

Tip: open the dashboard once before the review and press every preset.

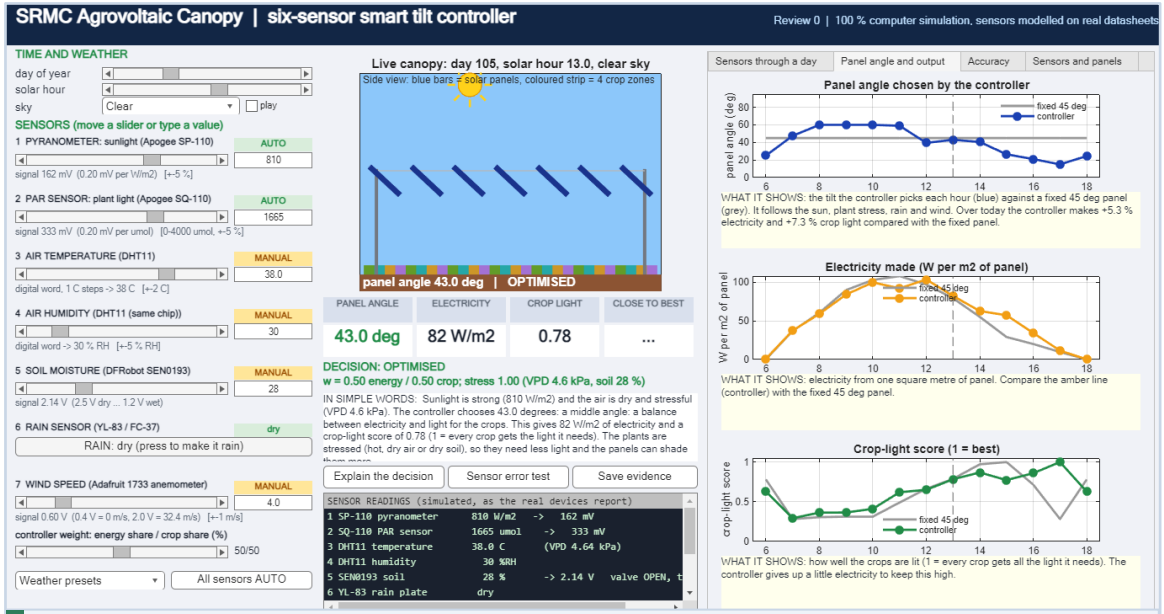
What the Graphs Show

Sensors through a day, and the panel angle and output against a fixed 45 deg panel



Sensors through a day

Sunlight and PAR rise and fall with the sun; temperature and humidity move together; soil dries until the valve opens; wind stays below the 15 m/s storm limit.

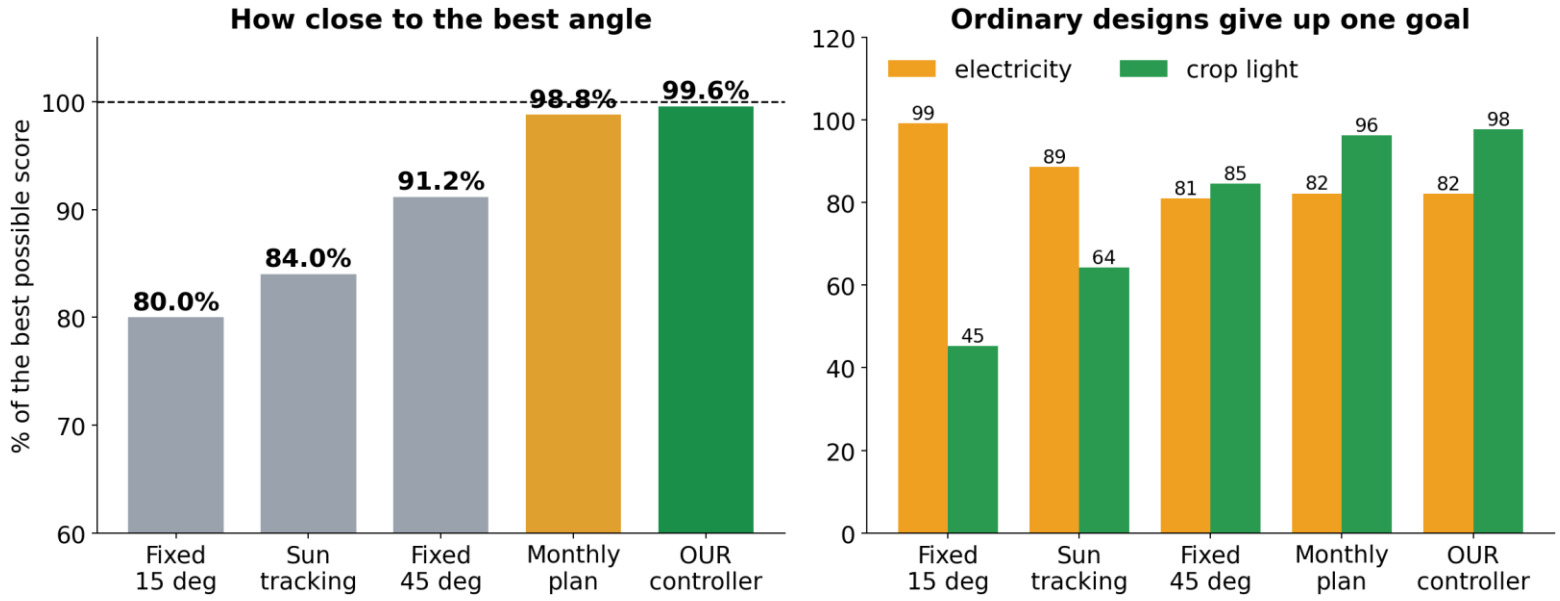


Angle and output

The controller changes the angle through the day (blue) while a fixed panel cannot. On this hot, clear April day it makes about +5 % electricity and +8 % crop light against fixed 45 deg.

Result: Accuracy of the Co-Optimisation

How close the chosen angle is to the best possible angle (half electricity, half crop light)



Design	Score	Electricity	Crop light	Error
Fixed 45 deg	91.2 %	81.0 %	84.6 %	16.3 deg
Our controller	99.6 %	82.1 %	97.7 %	4.8 deg

What we obtained

99.6 %

of the best possible score over 864 noisy test cases, with datasheet-sized sensor errors.

Ordinary designs reach 80 to 91 % and give up either crop light or electricity.

Simulation: measured against the controller's own goal, not real crop yield.

Testing and Validation

29 of 29 software tests pass; the sunlight model is checked against NASA POWER and PVGIS

Checks done

Review 0 tests: 16 of 16 pass (decision rules, sensor errors, physics, explanation)

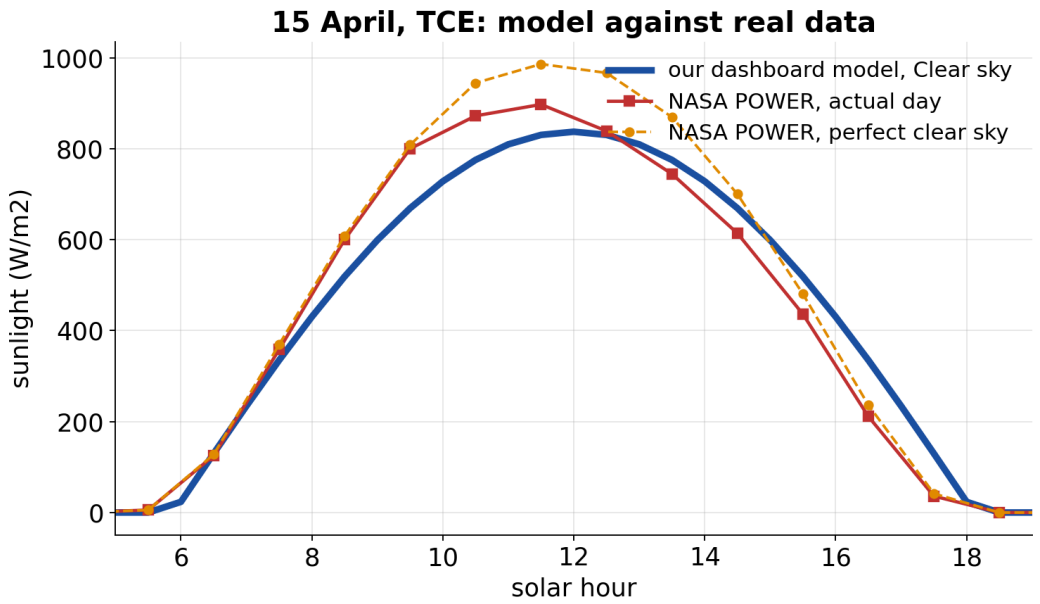
Sensor and tracker tests: 13 of 13 pass

Yearly electricity vs PVGIS: +5.0 %

Shadow model vs Blender 3-D: mean difference 0.010

Web twin and MATLAB model: identical results

Sunlight vs NASA POWER and PVGIS (right)



15 April	Wh/m2
Our Clear day	6502
NASA actual	6539
PVGIS mean	6337

Dashboard Clear day = a typical clear April day: daily energy within 0.6 % of NASA POWER actual day and 2.6 % above PVGIS April mean. Plant-light share: NASA 0.444, dashboard 0.45. Noon peak is below a perfectly cloudless day (837 vs 986 W/m2). These are satellite data, not a sensor on the TCE roof.

Proof: Datasheets, Data and Checks

Every file is in the Proof folder (Proof/10_Review0_Sensors_and_Panels) and on GitHub

What we can show	File	It proves
Pyranometer manual	01_pyranometer_ApogeeSP110_manual.pdf	0.20 mV per W/m2, +-5 %, 0-2000 W/m2
PAR manual	02_PAR_ApogeeSQ110_manual.pdf	0.20 mV per umol, +-5 %, 410-655 nm
DHT11 datasheet	03-04_airTemp_and_humidity_DHT11_datasheet.pdf	+-2 deg C, +-5 % RH
Soil and rain sensor pages	05_soilMoisture_*, 06_rain_*	output 1.2-2.5 V; wet/dry plate
Anemometer datasheet	07_wind_Adafruit1733_datasheet_C2192.pdf	0.4-2.0 V = 0-32.4 m/s
Panel datasheet	P1_panel_JinkoSolar_JKM550M-72HL4-V_datasheet.pdf	550 Wp, 40.90 V, 13.45 A, 21.33 %
NASA POWER and PVGIS data	sunlight_datasets/ (5 files)	where the sunlight data comes from
Model vs data	TRUST_sunlight_model_vs_NASA_PVGIS.png	daily energy within about 1-3 %
Accuracy test	T11_accuracy_v2.csv (864 cases)	99.6 % vs 91.2 % fixed 45 deg

GitHub

Also: SHA-256 checksum of every proof file (MANIFEST.csv) and the Review 0 Evidence Book (Word) for the judges.

Cost Estimate: If the Dashboard Sensors Were Real

Indicative estimates in INR, not quotes; the simulation itself costs nothing

Item	Research-grade kit	Low-cost kit
Sunlight sensor (Apogee SP-110 / reference cell)	28,000	1,500
PAR sensor (Apogee SQ-110 / reference cell)	28,000	1,500
Anemometer (Adafruit 1733)	6,500	6,500
DHT11 + soil sensor + rain sensor	700	700
ESP32 board + ADC (ADS1115 x 2)	1,400	1,400
Amplifier, enclosure, cables	3,000	1,500
Total sensing and control kit	**67,600**	**13,100**

Assumptions: Apogee sensors about US$ 250-300 each plus import duty and GST; other prices typical Indian retail. Replace with quotes before purchase. The earlier Phase II sensing kit (DHT11, soil, rain, ESP) was about INR 700. A year of calibration of the Apogee sensors is not included.

Panels (assumed)

8 x 550 Wp = 4,400 Wp

at about INR 16 per Wp:

about INR 70,000

This simulation

INR 0

MATLAB (college licence) and the web twin; no hardware.

Feasibility, SDG Relevance and Limits



SDGs

7 clean energy from farmland | 2 crops keep their light | 6 rain harvested for irrigation | 13 climate action | 8 farmer income

Feasibility

Sensing kit costs INR 13,100 to 67,600; all parts are standard and available in India

The controller needs only a small board; a distilled neural network for an ESP32 was tested in simulation

All results are reproducible: code and proof are on GitHub

Limits (we state them first)

Simulation only: sensor readings are generated or typed

Sunlight checked against satellite data, not a roof sensor

Accuracy is against our own goal, not crop yield

Prices and farmer share are assumptions

Review 0 Status, Remaining Work and Plan

What is finished, what is left for tonight, and what comes next

Done

- Dashboard with 6 real-model sensors, all manual
- Day graphs, accuracy and sensor-error tests
- 29 of 29 tests pass
- Datasheets and datasets downloaded as proof
- Evidence Book, poster, presentation
- Code on GitHub

Left for tonight

- Rehearse the 5-minute demo once
- Open the dashboard and press every preset
- Print the A3 poster
- Open the Jinko PDF and the SP-110 manual to be ready
- Charge the laptop, copy the Proof folder

Remaining after Review 0

- Real sensor set and one month of logged data
- Wind-tunnel or CFD run for the storm stow
- Crop-light and yield measurement
- Finish and submit the paper

Review 0
now: sensors and controller

Review 1
design and CAD

Review 2
solar and shadow

Review 3
digital twin

Review 4
control and simulation

Review 5
economics and final

Conclusion and Questions

Conclusion

Six sensors, each modelled on a real device and adjustable by hand, choose the panel angle live in a MATLAB dashboard

The co-optimisation reaches 99.6 % of the best possible score; ordinary designs reach 80-91 %

Sunlight is checked against NASA POWER and PVGIS; every datasheet and dataset is stored as proof

If built for real, the sensing kit costs INR 13,100 to 67,600

Next: real sensors, one month of data, wind check, paper

Likely questions

Are the sensors real? No: models of real devices, datasheets on file

Where does the sunlight come from? Sun position and NASA POWER clearness index, checked against NASA and PVGIS

Why trust 99.6 %? Best angle found by trying every angle, with sensor errors added

What is next? Real sensors and one month of data

Thank you | Questions and discussion